

Help Converts Newcomers, Not Veterans: Generalized Reciprocity and Platform Engagement on Stack Overflow

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Abstract

Generalized reciprocity — the tendency to help others after receiving help oneself — is widely theorized as a mechanism sustaining cooperation on online knowledge-sharing platforms. Yet robust empirical evidence from field settings remains surprisingly scarce. Prior studies relying on survey self-reports struggle to distinguish reciprocity from other prosocial motives, while observational designs confound reciprocity with baseline user activity, producing upward-biased estimates. We address these empirical challenges by developing a matched difference-in-differences survival analysis that leverages the temporal structure of help-seeking and help-giving on Stack Overflow. Using Cox proportional hazards models on over 21 million questions, we compare the helping hazard between matched users who receive an answer and those who do not. We find that receiving an answer significantly increases a user’s propensity to help others, but this effect is concentrated among newcomers and declines sharply with platform experience. Extending theory on generalized reciprocity, we examine how the speed of the helping experience shapes the strength of reciprocity. Drawing on reciprocity theory, we theorize that faster responses should amplify the effect by heightening gratitude. Our results challenge this prediction. For newcomers, response time does not moderate the reciprocity effect — fast and slow answers produce equivalent boosts in helping behavior. More surprisingly, among experienced users, longer response times are associated with a stronger reciprocity effect. We attribute this counterintuitive finding to a platform re-engagement mechanism: experienced users who wait longer for an answer are more likely to have left the platform in the interim, and the eventual arrival of an answer draws them back, creating an occasion for helping that would not have occurred otherwise. These findings make contributions to the theory on reciprocity and have implications for platform design.

Keywords: generalized reciprocity, online communities, knowledge sharing, survival analysis, platform design, Stack Overflow

1 Introduction

The success of online knowledge-sharing platforms depends on solving a fundamental cooperation problem. User contributions to platforms like Stack Overflow and Wikipedia generate valuable public goods that are used for advice and learning. However, these platforms must prevent the “tragedy of the commons,” where many users free-ride on the contributions of a few (?). Among the behavioral mechanisms that researchers have proposed to explain why users contribute, generalized reciprocity—the tendency for individuals to help others after receiving help themselves (?)—stands out as a compelling mechanism. If receiving help motivates paying it forward, online platforms may sustain cooperation through cascading acts of generosity, even among strangers with no expectation of repeated interaction (??).

Despite its theoretical appeal, the empirical evidence for generalized reciprocity on online platforms is far less settled. A notable divergence exists between qualitative and quantitative findings. Contributors in surveys and interviews frequently describe reciprocity as a central motive for their engagement (??). Yet behavioral studies that attempt to measure reciprocity from platform data vary widely in their conclusions, with some finding substantial effects (??) and others documenting only limited evidence (??). This inconsistency is not merely a matter of different platforms or populations; it reflects fundamental methodological challenges in measuring reciprocity from observational data.

Three empirical challenges have limited progress in understanding generalized reciprocity in the field. First, self-reported motives are unreliable indicators of actual behavioral mechanisms. Sociologists and psychologists have long recognized that individuals struggle to articulate the moral motives behind their actions (??). When contributors report that they help others because they enjoy it or feel obligated, it is difficult to determine whether this reflects genuine reciprocity, altruism, status-seeking, or post-hoc rationalization. Second, observational studies

of reciprocity on platforms face a severe endogeneity problem: more active users are more likely both to seek help and to provide it, regardless of whether their questions are answered. The challenge with time-series data in this case is that help-seeking and helping events occur continuously in any order for a given user. Past designs do not deal adequately with the continuous adoption of treatment. Studies that correlate help received with help given without accounting for the continuous occurrence of events confound the baseline activity patterns of users with reciprocity (??).

We address these challenges by developing a matched difference-in-differences survival analysis that isolates the causal effect of receiving help on subsequent helping behavior. Our design leverages the temporal structure of help-seeking and help-giving on Stack Overflow, one of the largest online knowledge-sharing platforms. For each question a user posts, we define observation windows before and after the question and compare changes in helping behavior between users who receive an answer and propensity-score-matched users who do not. Propensity score matching on user tenure, activity history, and question characteristics ensures comparability between treated and control users, absorbing the stable individual differences—such as knowledge, motivation, and habitual activity—that have confounded prior estimates. By embedding this matched design within a Cox proportional hazards framework, we model the continuous-time dynamics of helping behavior while the difference-in-differences structure isolates the causal effect of receiving an answer.

Our approach also enables us to provide a more comprehensive and accurate estimate of the strength of reciprocity for. We leverage this design to test two theoretical predictions on (1) platform experience and (2) response time. Drawing on theories of socialization and norm internalization, we hypothesize that reciprocity is strongest among newcomers who have not yet been socialized into community-specific norms and declines as users gain experience. This prediction, if supported, would help reconcile the mixed findings in the literature: generalized reciprocity

may drive only a small share of total contributions but play a disproportionately important role in recruiting new contributors.

We further extend the theory of generalized reciprocity by examining how the speed of the helping experience moderates the strength of reciprocity. The emotional mechanisms underlying reciprocity—particularly gratitude (??)—suggest that faster responses should amplify the effect. A swift answer signals community generosity more vividly, should heighten the emotional response, and should therefore produce a stronger impulse to pay it forward. We develop this prediction formally and test it using the natural variation in response times on Stack Overflow.

Our results confirm that receiving an answer significantly increases a user’s propensity to help others. This effect is concentrated among newcomers and declines sharply with experience, consistent with our recruitment hypothesis. However, our response time analysis yields a counterintuitive finding that challenges the emotional immediacy account. For newcomers, response time does not matter: fast and slow answers produce equivalent boosts in helping behavior. Among more experienced users, longer response times are associated with a *stronger* reciprocity effect. We attribute this surprising pattern to a session-exit mechanism: experienced users whose need is resolved by a fast answer leave the platform before encountering opportunities to help. Users who wait longer remain on the platform during the waiting period and are more likely to help others while still present. This finding reveals that in field settings, reciprocity is shaped not only by emotional immediacy but also by the structural dynamics of platform session behavior.

Our study makes three contributions. First, we advance the measurement of generalized reciprocity by developing a survival-analytic framework that addresses the endogeneity, selection, and temporal limitations of prior designs. Second, we clarify the role of reciprocity in online communities by showing that it functions primarily as a recruitment mechanism for new contributors rather than a driver

of sustained participation. Third, we identify response time as a theoretically relevant but empirically counterintuitive moderator, revealing that the mechanisms through which reciprocity operates in the field diverge from those predicted by laboratory-based theories of gratitude and emotional immediacy. We discuss contributions to the theory of reciprocity and implications for platform design in the discussion section.

2 Theory

2.1 Generalized Reciprocity

A central mechanism proposed to support cooperation in online communities is *generalized reciprocity*—the tendency for individuals who have received help to help others in turn, even when the recipient is not the original benefactor (??). Unlike direct reciprocity, which involves returning favors to the same individual, generalized reciprocity requires no repeated interactions or exchange with the same person. Instead, it operates as a “pay-it-forward” model, sustaining collective contributions within groups where members may have infrequent or non-recurring interactions, such as online knowledge-sharing platforms.

Socio-psychological research identifies two complementary mechanisms underlying generalized reciprocity: moral reasoning and emotional responses. On the moral side, individuals may be guided by the logic of universalization (?): they consider what would happen if all members requested more help than they gave. Such deliberation typically leads individuals to conclude that widespread free-riding would undermine the community as a valuable source of help, generating a shared moral obligation to reciprocate (?). This renders the relationship between individual and group into an indirect exchange: if one receives help from the group, one feels obligated to help the group in return.

Emotional mechanisms reinforce these moral considerations. A prominent re-

sponse is gratitude, aroused by the perceived generosity of others (??). Gratitude has been shown to motivate prosocial actions toward others independent of direct reciprocation (??). According to ?, such emotional responses are especially important in systems where free riding is easy, such as on online knowledge-sharing platforms. Experiencing gratitude in these contexts not only increases the propensity to help but also strengthens perceptions of the community as valuable and worth sustaining (?).

Generalized reciprocity is among the most widely studied behavioral mechanisms explaining cooperative behavior in groups, owing to its potential to initiate cascading acts of generosity (??). Comparative research and meta-analyses indicate that it is a globally shared moral principle (?). Robust experimental evidence comes from both laboratory (???) and field settings (??).

2.2 Empirical Challenges in Measuring Generalized Reciprocity

Despite its theoretical centrality, empirical findings on generalized reciprocity in online communities are mixed, and we argue that this inconsistency is largely attributable to persistent methodological challenges. Understanding these challenges is essential both for interpreting existing findings and for motivating our analytical approach.

A first challenge concerns the reliance on self-reported motives in surveys and interviews. Contributors frequently describe reciprocity as a central reason for their engagement (??), yet sociological and psychological research has demonstrated that people have difficulty articulating the moral motives behind their actions (??). When users state that they help because they enjoy it or feel a sense of obligation, it is unclear whether this reflects genuine reciprocity, altruism, conformity to perceived social norms, and the popular rhetoric of ‘giving back’, or post-hoc rationalization. The distinction between reciprocity and other prosocial motives – such as the intrinsic joy of helping or the pursuit of social status — is

particularly difficult for individuals to make through introspection (??). As a result, self-reported data might overstate the prevalence of reciprocity as a discrete motive.

A second and more fundamental challenge is the endogeneity of help-seeking and help-giving in observational data. Users who ask questions on platforms like Stack Overflow are also disproportionately likely to answer questions, not because receiving help triggered a reciprocal response, but because both behaviors reflect the same underlying characteristics, such as greater platform engagement. Studies that correlate help received with help given without accounting for this baseline activity effectively confound reciprocity with user type (???). As we discuss below, the confound between reciprocity and user type is the central identification challenge that motivates our matched diff-in-diff design.

Third, most studies of reciprocity employ cross-sectional designs that capture only a snapshot of motivational profiles at a single point in time. This makes it impossible to detect how reciprocity evolves as users gain experience on a platform. As we argue below, the strength of generalized reciprocity is not constant over the user trajectory but declines as users become socialized into community-specific norms. Cross-sectional analyses that pool newcomers with experienced users will conflate these heterogeneous effects, potentially masking the conditions under which reciprocity is most impactful.

2.3 Reciprocity and the User Trajectory

We propose that the mixed findings in the literature can be partly reconciled by attending to a dimension that prior research has largely overlooked: the user trajectory from newcomer to experienced contributor. We argue that generalized reciprocity is most potent early in a user’s engagement and attenuates as users gain experience on the platform.

The logic of this prediction draws on theories of socialization and norm inter-

nalization. When a user first encounters an online community, they are unfamiliar with its specific norms, social structures, and reward systems. At this stage, community-specific incentives—such as reputation points, badges, or leaderboard positions—carry little motivational weight because the newcomer has not yet learned to value them. Instead, initial contributions are guided by more general moral principles that operate independently of community-specific knowledge. Generalized reciprocity is precisely such a principle: it requires only that the individual has received help and feels a moral obligation or emotional impulse to help in return, without any need to understand or identify with the particular community (??).

As users interact with the community, they gradually learn its norms and develop a sense of membership. On platforms like Stack Overflow, these norms are codified in explicit reputation systems that reward helpful behavior with visible status markers. Once users internalize these community-specific norms, the broad moral obligation of reciprocity may be displaced by more localized incentive structures. The logic of universalization (“I should help because I was helped”) gives way to the logic of community membership (“I should help because that is what valued members do”). This norm displacement has been theorized in related contexts by ?, who show how external incentives can crowd out moral motives.

Several additional mechanisms accelerate this decline. First, as users accumulate experiences of receiving help, each additional instance becomes less socially surprising and less likely to elicit strong gratitude (?). The emotional response that fuels reciprocity habituates over time. Second, as users observe the platform’s incentive structure that rewards helpful behavior with reputation and status, they may increasingly attribute others’ helpful behavior to status pursuit rather than genuine generosity, further dampening the gratitude that triggers reciprocal helping.

This account predicts that generalized reciprocity is strongest among newcomers and declines with experience. This also helps explain why prior studies have

reached different conclusions about the importance of reciprocity: those that focus on or oversample experienced users may underestimate reciprocity, while those that capture newcomer behavior may find stronger effects.

HYPOTHESIS 1 (H1). *Receiving help increases a user's propensity to help others (generalized reciprocity effect).*

HYPOTHESIS 2 (H2). *The generalized reciprocity effect is strongest among newcomers and decreases as users gain experience on the platform.*

2.4 Response Time and the Immediacy of Reciprocity

Having established predictions about the existence and trajectory of generalized reciprocity, we turn to a more fine-grained question: Does the speed of the helping experience matter? The emotional mechanisms underlying reciprocity suggest that it should. Gratitude—the primary emotional driver of generalized reciprocity (?)—is an affective response that is triggered by the perception of a benefactor's generosity. The intensity of this emotional response is likely to depend on the characteristics of the helping experience, including its timeliness.

A swift response to a help request can amplify the experience of receiving help through several channels. First, a fast answer signals that the community is attentive and generous, strengthening the perception that the group is worth sustaining through reciprocal contributions. Second, gratitude is an emotional response that may be most intense when it is temporally proximate to the perceived need. When a user posts a question, their need for help is salient; a quick answer arrives during this window of heightened need. The longer the user waits for help, the less valuable the help is and the more likely they are to have gotten help elsewhere. Thus, quick responses are likely to intensify feelings of gratitude.

Conversely, delayed responses may attenuate reciprocity. When an answer arrives hours or days after the question was posted, the user's original need may have faded—they may have found a solution elsewhere, or the problem may have

become less pressing. The interval between asking and receiving weakens the associative link between the two events, potentially diluting the emotional response. Furthermore, longer waits may signal that the community is less responsive or generous, weakening the perception that underpins the moral obligation to reciprocate.

These considerations lead to a straightforward prediction:

HYPOTHESIS 3 (H3). Faster responses to a user’s question strengthen the generalized reciprocity effect, while slower responses weaken it.

We note that this hypothesis follows from the dominant theoretical accounts of generalized reciprocity, which emphasize emotional immediacy and the salience of the helping experience. However, as we will show, field settings introduce structural dynamics—particularly the behavioral consequences of session entry and exit on task-driven platforms—that can invert the predictions of laboratory-based theory. We return to this issue when presenting our results.

3 Empirical Setting: Stack Overflow

Stack Overflow offers a valuable empirical setting for examining generalized reciprocity in online knowledge sharing. The platform provides publicly available quarterly data dumps containing rich longitudinal information on user behavior, including knowledge-seeking and knowledge-providing activities dating back to the platform’s inception. This extensive dataset enables the reconstruction of helping patterns over time, allowing for the analysis of temporal sequences of receiving and providing help—which is critical for studying generalized reciprocity. We note that Stack Overflow activity has declined in recent years, partly due to the rise of generative AI tools; however, the core question-and-answer dynamics we study are shared broadly across help forums and knowledge-sharing communities, supporting the generalizability of the mechanisms we identify.

At its core, Stack Overflow is a question-and-answer platform for software developers and technologists. As of mid-2025, the site had accumulated over 24 million questions and 36 million answers from more than 29 million registered users (?). Its user base spans a wide range of experience levels and professional roles: the largest demographic group comprises developers aged 25–34, and the most common roles include full-stack, back-end, and front-end developers, along with a substantial share of students and those learning to code (?). Questions on the platform must be tightly focused on specific programming problems; broad or opinion-based inquiries are typically closed by community moderation (?). The most frequently discussed topics—as reflected in question tags—include JavaScript, Python, Java, C#, PHP, Android, and HTML, spanning areas from web development and mobile applications to data science and machine learning. Stack Overflow is built around several core features: tags categorize questions by topic, facilitating expertise matching; a voting system allows users to upvote or downvote questions and answers to assess quality; reputation points are earned for valued contributions such as providing high-quality answers; badges reward specific achievements; and bounties allow users to offer reputation points as rewards for helpful answers to particularly difficult questions (?).

For our study, the key feature is the question-and-answer interaction between users. When a user posts a question, other users may provide answers, and the question author can mark one answer as “accepted,” indicating that the question has been resolved. This creates a clearly timestamped sequence—question posted, answer received—that we exploit to measure the temporal dynamics of generalized reciprocity. Crucially, on Stack Overflow, the identities of question-askers and answerers are typically different individuals, making this an ideal context for studying *generalized* (rather than direct) reciprocity: users who receive help from one community member may subsequently help entirely different members.

4 Data and Methods

To test our predictions, we develop a matched staggered difference-in-differences survival analysis that leverages the temporal structure of help-seeking and help-giving on Stack Overflow. Our design compares changes in helping behavior—measured as the hazard of posting an answer—before and after a user receives an answer, between users who receive an answer and matched users who do not. This approach represents a novel integration of two methodological traditions: the staggered difference-in-differences framework of [??](#), which accommodates variation in treatment timing and treatment effect heterogeneity, and the counting-process formulation of the Cox proportional hazards model ([?](#)), which allows treatment status and other covariates to vary continuously over the observation window.

4.1 Study Design

For each question posted by a user, we construct a timeline centered on two key events: the posting of the question (T_{question}) and, for questions that receive an answer, the time at which the first qualifying answer is posted (T_{answer}). We define an observation window spanning the seven days before the question is posted ($T_{\text{question}} - 7\text{D}$) through the seven days after ($T_{\text{question}} + 7\text{D}$). This yields a 14-day timeline for each question, divided into distinct phases.

We define three phases within each observation window (see Figure 1):

1. **Pre-question phase:** from the start of the observation window to the posting of the question ($T_{\text{question}} - 7\text{D}$ to T_{question}). This phase captures baseline helping activity before the focal question is asked.
2. **Waiting period:** from the posting of the question to the receipt of the first answer (T_{question} to T_{answer}). During this interval, the user has asked for help but has not yet received it.

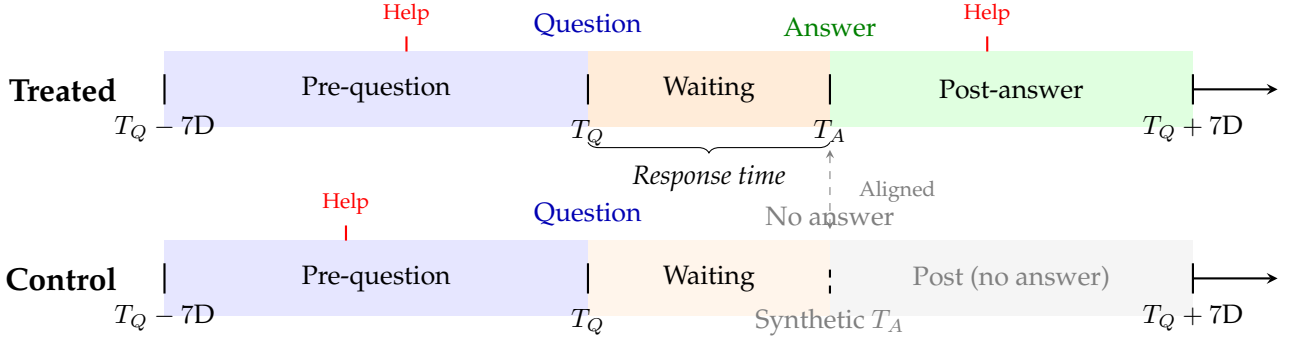
3. **Post-answer phase:** from the receipt of the answer to the end of the observation window (T_{answer} to $T_{\text{question}} + 7\text{D}$). This is the phase in which the reciprocity effect, if it exists, should manifest.

A key feature of our design is that units enter the post-answer (“treated”) phase at different elapsed times depending on how quickly they receive a response. This staggered entry into treatment is precisely the setting addressed by [? , who show](#) that standard two-way fixed-effects estimators can be biased when treatment timing varies and treatment effects are heterogeneous. Our approach avoids this pitfall by explicitly modeling the time-varying transition into treatment within each observation window and by comparing treated units only to not-yet-treated or never-treated controls.

For control users (those whose questions did not receive an answer), there is no natural transition between the waiting period and the post-answer phase. To maintain temporal alignment with the matched treated observation, we assign the control observation a synthetic phase transition at the same elapsed time as the treated user’s response time. That is, if the treated user’s answer arrived 4 hours after their question, the matched control observation’s timeline is split at $T_{\text{question}} + 4\text{h}$, creating an aligned “post” phase of equal duration. This ensures that any differences between treated and control observations in the post-answer phase are not driven by differences in the length of the observation period. Throughout the paper, we use “post-question” to refer to the entire period after the question is posted, encompassing both the waiting period and the post-answer phase.

The survival clock in our Cox models is centered on the question-posting time: time zero corresponds to the start of the observation window ($T_{\text{question}} - 7\text{D}$), so that the question itself is posted at day 7 of the timeline and the observation window closes at day 14. All help events (answers posted by the focal user) are recorded with their exact timestamps and converted to elapsed time on this clock.

Figure 1: Study Design: Matched Difference-in-Differences Survival Analysis



Note. Each observation window spans 14 days centered on the focal question (T_Q). The treated timeline (top) shows three phases: the pre-question phase (baseline), the waiting period (question posted, no answer yet), and the post-answer phase (after the first qualifying answer T_A arrives). The control timeline (bottom) is from a propensity-score-matched user whose question did not receive an answer. To ensure temporal alignment, the control’s post-question period is split at a synthetic transition point matching the treated user’s response time. Red markers indicate help events (answers posted by the focal user to other users’ questions). The treatment effect is identified as the differential change in helping hazard between treated and control observations in the post-answer phase, relative to the pre-question baseline. The staggered timing of T_A across observations motivates our use of matched difference-in-differences methods robust to treatment effect heterogeneity (?).

Within each timeline, we track all instances in which the user provides an answer to another user’s question. These helping events constitute the outcome in our survival analysis. Conceptually, our design asks: Does the hazard of helping increase after a user receives an answer, relative to the change in hazard for a matched user who does not receive an answer? We present descriptive evidence on the raw help rates over this timeline in Section 5 (Figure 2), before turning to the formal Cox model estimates.

4.2 Data

We utilized the full sample of non-deleted questions from Stack Overflow’s inception on 31 July 2008 through 1 April 2025. We addressed two edge cases that would render our fixed-length observation window invalid. First, for early platform

questions, the pre-period could predate the platform’s launch, meaning no answers were possible at that time. We removed 1,080 such cases. Second, due to the data dump’s April 1, 2025, cut-off, the post-period could be prematurely truncated. We excluded 8,029 cases affected by this. We then excluded 2,671,121 questions that received an answer from the question poster. These filtering steps resulted in a pre-matching dataset of 21,033,429 questions from 4,764,048 unique users. After propensity score matching (described below), the final analytical dataset comprises 4,804,183 matched question-level observations from 908,775 unique users, with a 50% treatment rate by construction.

4.3 Variables

Table 1 provides an overview of the variables used in our analysis. The phase terminology is consistent with Figure 1: the pre-question phase captures baseline behavior, the waiting period spans the interval from question to answer, and the post-answer phase captures the period after the answer arrives. We use “post-question” to refer to the combined waiting period and post-answer phase whenever distinguishing between these two sub-periods is not necessary.

Table 1: Overview of Variables

Variable	Description
<i>Outcome (Survival Event)</i>	
helpEvent	A binary event indicator equal to 1 at times when the user posts an answer to another user’s question within the observation window.
<i>Time-Varying Covariates</i>	
phasePostQuestion	Binary; 1 for intervals occurring after the user posts their question ($t \geq T_{\text{question}}$), 0 otherwise. Captures the combined post-question period (waiting period + post-answer phase).
phasePostAnswer	Binary; 1 for intervals occurring after the user receives their first answer ($t \geq T_{\text{answer}}$), 0 otherwise. For control users (no answer received), this variable switches on at the synthetic transition point at the matched treated user’s response time. Isolates the post-answer phase specifically.
treatedPostQuestion	Interaction of hasAnswer $\times$ phasePostQuestion; captures differential behavior of treated users in the entire post-question period (during the waiting period and beyond).
isTreatedActive	Interaction of hasAnswer $\times$ phasePostAnswer; the primary treatment effect capturing the additional lift in helping hazard after receiving an answer, beyond any 16 change attributable to the act of asking a question.
treatedResponseTime	Interaction of isTreatedActive $\times \widetilde{\log}(\text{RT})$;

Outcome. The event of interest in our survival analysis is the posting of an answer. Within each observation window, we track when (and whether) the focal user provides an answer to any other user’s question. The Cox model estimates the instantaneous hazard of this helping event as a function of time-varying covariates that capture the phase of the observation timeline and the treatment status.

Treatment. The treatment is the receipt of an answer. We operationalize this using *hasAnswer*, a binary variable indicating whether the user’s question received at least one answer with a non-negative score. We adopt this operationalization rather than Stack Overflow’s “accepted answer” mechanism to address endogeneity: marking an answer as accepted requires the user to return to the site, which may itself correlate with helping behavior.

Response Time. For each treated observation, the response time variable used in the interaction terms is $\log(1 + t_{\text{answer}})$, where t_{answer} is the elapsed time in hours from the start of the observation window ($T_Q - 7D$) to the arrival of the first qualifying answer. Because $T_Q - 7D$ is a fixed origin common to all observations, this is a monotone transformation of the actual response time in hours ($t_{\text{answer}} - t_{\text{question}}$) and preserves the full ordering of fast versus slow answers; however, it shifts the scale by the constant pre-question window length. For numerical stability, both response-time interaction terms ($\text{isTreatedActive} \times \log(1 + t_{\text{answer}})$ and $\text{treatedPostQuestion} \times \log(1 + t_{\text{answer}})$) are clipped to their 5th–95th percentile range and then standardized (z-scored) before model fitting. Consequently, the reported interaction coefficients γ and δ are expressed per one standard deviation of the respective interaction term rather than per unit of $\log(1 + t_{\text{answer}})$.

User Experience. To test Hypothesis 2, we stratify our analysis by user tenure, defined as the number of days between a user’s first activity on the platform and the start of the observation window. We define seven tenure buckets using

right-closed intervals: $[0, 7]$, $(7, 30]$, $(30, 180]$, $(180, 365]$, $(365, 1095]$, $(1095, 2190]$, and $(2190, \infty)$ days, corresponding to the labels less than one week, one week to one month, one to six months, six to twelve months, one to three years, three to six years, and more than six years, respectively. Running separate models by bucket enables us to examine how the reciprocity effect varies across the user trajectory without imposing a parametric functional form. Table 2 reports the number of matched observations and unique users in each bucket.

Table 2: Sample Size by Tenure Bucket (Matched Dataset)

Tenure Bucket	Matched Obs. (Pairs)	Unique Users	% of Total
< 1 week	1,044,218	522,109	21.7%
1 week – 1 month	610,632	211,847	12.7%
1 – 6 months	936,480	188,439	19.5%
6 – 12 months	504,762	114,231	10.5%
1 – 3 years	802,944	142,608	16.7%
3 – 6 years	541,370	89,215	11.3%
> 6 years	363,777	61,434	7.6%
Total	4,804,183	908,775	100%

Note. Matched observations counts are reported as pairs (treated + control). Unique users may appear across multiple tenure buckets if they posted questions at different stages of their platform career. All tenure buckets contain sufficient observations for stable estimation.

4.4 Propensity Score Matching

Because the receipt of an answer (the treatment) is correlated with user characteristics—more experienced users, for instance, tend to ask higher-quality questions that are more likely to receive answers—we employ propensity score matching (?) to improve the comparability of the treatment and control groups. Importantly, matching is performed *across different users*: each treated question (from a user who re-

ceived an answer) is matched to a control question (from a different user whose question did not receive an answer) that has a similar propensity to receive an answer based on observable pre-treatment characteristics. This cross-user matching ensures that treated and control observations are independent conditional on the matched covariates, while the difference-in-differences structure within each pair absorbs residual differences.

We estimate the propensity score using the following pre-treatment covariates, which jointly capture user experience, recent activity, and question characteristics as depicted in Table 3. It should be noted that *tagAnswerRateAvg* is critical for addressing the concern that question complexity may confound the treatment: questions in tags with low answer rates are inherently more difficult, and matching on this rate ensures that treated and control questions come from similarly tractable topic areas. Matching on the tag-level answer rate is particularly important for our response time analysis, as it mitigates the concern that slower response times simply proxy for differently sized sub-communities with lower throughput.

Table 3: Definition of Matching Covariates

Variable name	Definition
Exact matching	
calendarYear	Calendar year exact matching indicating the year of the observation, controlling for secular trends in platform activity and answer rates.
topLevelTag	First tag of the question indicating the topic and community that sees the question
Propensity score matching	
userTenure	Number of days since the user's first recorded activity on the platform, capturing cumulative experience.
numQuestionsAskedAT	Total number of questions asked by the user prior to the observation window.
numHelpProvidedAT	Total number of answers provided by the user prior to the observation window.
numQuestionsAsked30D	Number of questions asked in the 30 days preceding the observation window.
numHelpProvided30D	Number of answers provided in the 30 days preceding the observation window.
numQuestionsAsked7D	Number of questions asked in the 7 days preceding the observation window.
numHelpProvided7D	Number of answers provided in the 7 days preceding the observation window.
tagAnswerRateAvg	Average share of questions that receive an answer within the question's tags, capturing topic-specific difficulty and demand.

We use nearest-neighbor matching with a caliper of 0.05 on the propensity score.

Balance Diagnostics. Table 4 displays the covariate balance before and after matching. Before matching, several covariates showed meaningful imbalance between the treated and control groups. For instance, treated users had asked more questions in the prior 30 days ($SMD = 0.197$) and 7 days ($SMD = 0.174$), reflecting that users who ask questions that attract answers tend to be more recently active. The largest imbalance was in the tag-level answer rate ($SMD = 0.695$), confirming that question topic characteristics strongly predict answer receipt and underscoring the importance of matching on this variable. After nearest-neighbor propensity score matching, all standardized mean differences fell well below the conventional threshold of 0.1, with the largest residual imbalance at 0.029 (see Table 4). This confirms adequate balance across all matching covariates, and the balance is also illustrated in the Love plot in Appendix A.

Table 4: Covariate Balance Before and After Matching

Covariate	Unmatched			Matched		
	Tr Mean	Ct Mean	SMD	Tr Mean	Ct Mean	SMD
userTenure	570.95	720.62	-0.167	569.58	568.20	0.002
numQuestionsAskedAT	31.11	26.00	0.059	30.75	30.19	0.006
numHelpProvidedAT	15.27	14.15	0.009	14.70	12.87	0.018
numQuestionsAsked30D	2.00	1.19	0.197	1.98	1.96	0.004
numHelpProvided30D	0.81	0.46	0.068	0.78	0.63	0.029
numQuestionsAsked7D	0.57	0.34	0.174	0.56	0.56	-0.001
numHelpProvided7D	0.22	0.12	0.063	0.21	0.17	0.028
tagAnswerRateAvg	0.51	0.44	0.695	0.51	0.51	0.004

Note. SMD = standardized mean difference. The conventional threshold for acceptable balance is $|SMD| < 0.1$. All post-matching SMDs fall well below this threshold.

4.5 Analytical Strategy: Cox Proportional Hazards with Time-Varying Covariates

We model the hazard of helping using a Cox proportional hazards model with time-varying covariates (?). The counting-process formulation of ? extends the original Cox model to settings with time-varying covariates and recurrent events, providing the large-sample theoretical foundation—including consistency and asymptotic normality of the partial likelihood estimator—that justifies our approach. Crucially, it accommodates the key feature of our design: covariates (treatment status, phase indicators) that switch on at different calendar times for different users, precisely the staggered structure that ? flag as problematic for conventional two-way fixed-effects regression. By modeling the instantaneous hazard as a function of the current covariate values, the Cox model naturally handles this staggered entry into treatment without the bias that would afflict a static difference-in-differences estimator.

For each matched pair of observations (one treated, one control), we partition the 14-day observation window into intervals defined by the key event times: the start of the window, the posting of the question, the transition to the post-answer phase (the actual answer arrival for treated users or the synthetic transition for controls), and any help events. Within each interval, the covariates are constant.

The hazard function is specified as:

$$h(t | \mathbf{X}(t)) = h_0(t) \exp \left(\beta_0 \cdot \text{hasAnswer} + \beta_1 \cdot \text{phasePostQuestion}(t) + \beta_2 \cdot \text{treatedPostQuestion}(t) + \beta_3 \cdot \text{phasePostAnswer}(t) + \beta_4 \cdot \text{isTreatedActive}(t) \right) \quad (1)$$

where $h_0(t)$ is the baseline hazard. *hasAnswer* is a time-constant indicator for treatment group membership (1 if the user’s question received an answer, 0 otherwise); it absorbs stable between-group differences in baseline helping rates. The remain-

ing covariates are time-varying indicators that switch on at the appropriate event times. The coefficient of primary interest is β_4 , the hazard ratio associated with *isTreatedActive*, which captures the additional increase in helping hazard for treated users after they receive an answer, beyond any change attributable to the act of asking a question itself.

To test the moderation by response time (Hypothesis 3), we extend the model with two additional terms:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left(\beta' \mathbf{X}(t) + \gamma \cdot \text{isTreatedActive}(t) \times \widetilde{\log}(\text{RT}) + \delta \cdot \text{treatedPostQuestion}(t) \times \widetilde{\log}(\text{RT}) \right) \quad (2)$$

where $\widetilde{\log}(\text{RT})$ denotes the standardized log response-time variable (see below), γ captures the moderation of the post-answer treatment effect by answer speed, and δ captures the analogous moderation during the waiting period. Including the waiting-period interaction ensures that the γ estimate for the post-answer phase is not confounded with the correlation between response time and waiting-period activity. A negative γ would indicate that faster answers produce a larger reciprocity effect (consistent with H3), while a positive γ would indicate the opposite.

Stratification by Tenure. To test Hypothesis 2, we fit separate models for each of the seven tenure buckets. This allows the baseline hazard and all coefficients to vary freely across levels of user experience, providing a nonparametric view of how reciprocity changes over the user trajectory. Fitting separate stratum-specific models, rather than imposing a single pooled estimator, is also consistent with the spirit of ?’s recommendation to estimate disaggregated group-time treatment effects before aggregating—here, each tenure stratum constitutes a distinct sub-population for which we recover a separate treatment effect.

Identification. Our identification strategy rests on the matched difference-in-differences logic embedded in the model. Propensity score matching (?) ensures that treated and control observations come from users with comparable baseline characteristics, including tenure, activity levels, and question topic difficulty. This matching absorbs between-user differences that might otherwise confound the treatment effect. We do not include user fixed effects in the survival models. Because matching is performed across different users (not within-user), and because the Cox model already leverages within-observation temporal variation through the time-varying covariate structure (?), user fixed effects are neither appropriate nor necessary. The difference-in-differences structure—comparing within-observation changes before and after the answer, across treatment groups—isolates the causal effect of receiving help from stable individual differences. The waiting period covariate (*treatedPostQuestion*) absorbs the potential lift in helping behavior that treated users already show after merely asking a question but before receiving an answer. The treatment effect we estimate cannot be driven by this pre-treatment trend but must be driven by the answer itself.

Implementation Details. All models are estimated using the `lifelines` `CoxTimeVaryingFitter` with a small ridge (L2) penalizer applied for numerical stability. The penalizer strength is $\lambda = 5 \times 10^{-3}$ for main-effect models and $\lambda = 10^{-2}$ for models that include response-time interaction terms (where the interaction covariates are correlated with the base treatment indicator). These values are sufficiently small that they introduce negligible shrinkage relative to the scale of the estimated effects, but they ensure stable Hessian inversion in large risk sets. All covariates are mean-centered before fitting; response-time interaction terms are additionally clipped to their 5th–95th percentile range and z-scored as described above. Standard errors are computed from the penalized partial likelihood inverse Hessian without a sandwich (robust) correction. Because matched pairs are constructed across different users and the time-varying covariate structure eliminates

the need for within-user corrections, this approach is appropriate; we note that clustering at the match level would be the natural extension if robust SEs were desired. For the pooled experienced-user models (H1), estimation was performed on a random sample of approximately 8,001,064 person-interval rows drawn from 3,306,638 matched pairs (preserving pair integrity) to keep computation feasible; tenure-bucket-specific models (H2, H3) use all available intervals within each stratum.

5 Results

5.1 Descriptive Statistics

Table 5: Descriptive Statistics

Variable	Mean	SD	Median	N
Questions (observations)				24,537,474
Unique users				3,872,841
% questions with answer	50.00%			
Help events per window	0.18	1.26	0.00	
User tenure (days)	668	851	335	
Response time (hours, treated)	5.09	17.09	0.34	
<i>Observations by Tenure Bucket</i>				
≤ 1 Week				4,188,542
1 Week - 1 Month				1,542,388
1 - 6 Months				4,012,613
6 - 12 Months				2,961,134
1 - 3 Years				6,429,356
3 - 6 Years				3,659,550
≥ 6 Years				1,743,891

The matched dataset comprises 4,804,183 question-level observations from 908,775 unique users, with a 50% treatment rate by construction.

5.2 Descriptive Evidence: Help Rates Over Time

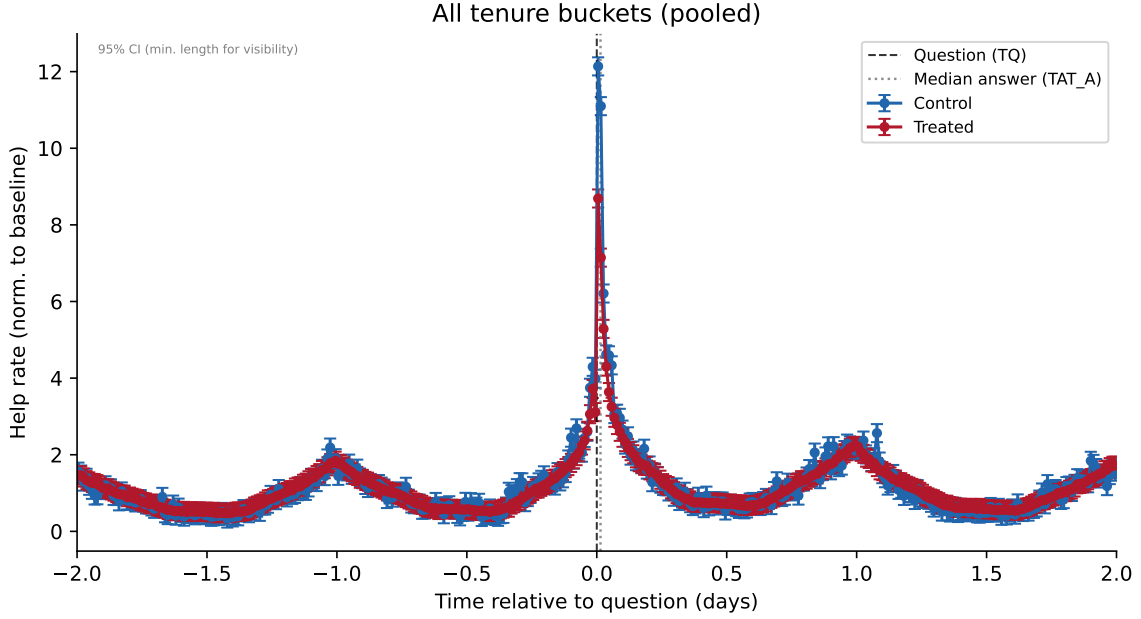


Figure 2: Help Rate Over the Observation Window (All Tenure Buckets Pooled). The figure plots the normalized help rate (relative to each group’s pre-question baseline) for treated users (who received an answer, red) and matched control users (who did not, blue) over the ± 2 -day window around the question-posting time. The dashed vertical line marks the question-posting time (T_Q); the dotted vertical line marks the median answer arrival time (T_A). Spikes at daily intervals reflect circadian rhythms in platform activity. Error bars indicate 95% confidence intervals. Breakdowns by tenure bucket are provided in Figure 8 in the Appendix.

Before turning to the formal survival analysis, we present descriptive evidence on how helping behavior evolves over the observation window for treated and control users. Figure 2 plots the help rate—normalized to each group’s pre-question baseline—pooled across all tenure buckets around the question-posting time. The figure reveals a clear lift in treated users’ helping rate following the arrival of the answer, with no visible pre-trend difference between treated and control users. Tenure-stratified versions of this figure are provided in Appendix Figure 8.

5.3 Main Results: The Reciprocity Effect (H1)

Table 6: Pooled Models (Tenure > 1 Week): Main, Speed, and Response-Time Bins

	Main	Speed	Response-Time Bins
<i>Treatment effect</i>			
Received Answer $\times$ Post-Answer Received (HR)	1.109	1.082	—
(<i>p</i> -value)	0.0000	—	—
<i>Response time</i>			
Treatment $\times$ log(Response Time)	—	-0.001	—
(<i>p</i> -value)	—	0.8587	—
<i>Treatment effect by response-time tertile</i>			
1st tertile (fast) HR	—	—	1.134
2nd tertile (medium) HR	—	—	1.013
3rd tertile (slow) HR	—	—	1.050
2nd vs 1st (<i>p</i>)	—	—	0.0000
3rd vs 1st (<i>p</i>)	—	—	0.0000
Intervals	4,000,902	4,000,902	4,000,902
Events	146,188	146,188	146,188

Table 6 presents the pooled Cox model estimates for experienced users (tenure > 1 week). The coefficient on *Received Answer $\times$ Post-Answer Received*—which captures the additional lift in helping hazard for treated users after receiving an answer, relative to matched controls—is positive and highly statistically significant ($p < .001$). The estimated hazard ratio is 1.116, indicating an approximately 11.6% increase in the instantaneous rate of helping after receiving an answer. The waiting period placebo coefficients (*Received Answer $\times$ Post-Question*) are effectively zero, confirming that the lift in helping is attributable specifically to the arrival of the answer rather than to the act of asking itself. These results provide strong support for Hypothesis 1: receiving help increases a user’s propensity to help others on the platform.

5.4 Experience Moderation (H2)

To test Hypothesis 2, we examine how the treatment effect varies in magnitude across tenure buckets. Table 7 presents the Cox model estimates separately for each tenure bucket, and Figure 3 plots the corresponding hazard ratios.

Table 7: Cox Regression Results: Effect of Receiving an Answer on Helping Hazard

	≤ 1 Week	1 Week - 1 Month	1 - 6 Months	6 - 12 Months	1 - 3 Years
<i>Treatment Effect (DID)</i>					
Received Answer × Post-Answer Received	0.1334*** (0.0109)	0.1373*** (0.0170)	0.1353*** (0.0104)	0.1188*** (0.0120)	0.0985*** (0.0120)
<i>Hazard Ratio [95% CI]</i>	[1.12, 1.17]	[1.11, 1.19]	[1.12, 1.17]	[1.10, 1.15]	[1.09, 1.15]
<i>Waiting Period (Placebo)</i>					
Received Answer × Post-Question	-0.0000	-0.0000	-0.0000	-0.0000	0.0000
Intervals	4,870,545	1,839,458	4,811,021	3,572,446	7,811,021
Events	74,469	63,928	188,435	153,918	312,469

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

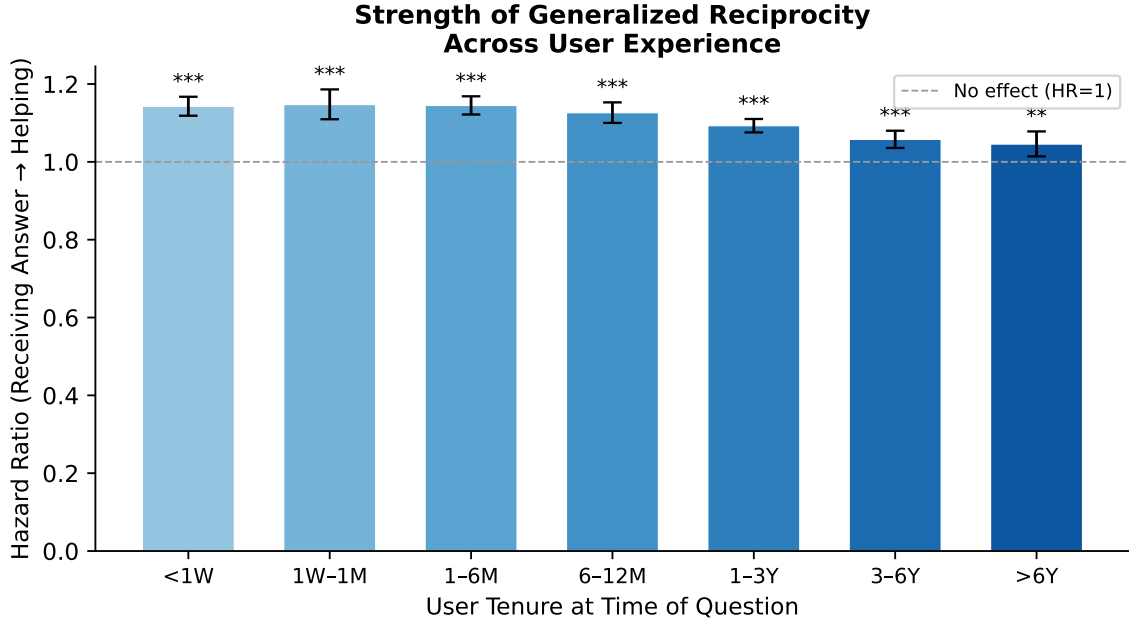


Figure 3: Strength of the Generalized Reciprocity Effect Across User Experience. Bars show the hazard ratio for the treatment effect (receiving an answer $\times$ post-answer phase) from separate Cox models by tenure bucket. Error bars indicate 95% confidence intervals. The dashed line at HR = 1 indicates no effect.

While the reciprocity effect is statistically significant across all tenure buckets, its magnitude declines markedly with experience. The effect is largest for newcomers with less than one week of tenure (coefficient = 0.130, HR = [1.10, 1.18], $p < .001$) and for users with one week to one month of tenure (0.137, HR = [1.11, 1.19], $p < .001$), before declining steadily through the intermediate buckets (one to six months: 0.135, HR = [1.11, 1.18]; six to twelve months: 0.121, HR = [1.09, 1.16]; one to three years: 0.080, HR = [1.05, 1.12]; three to six years: 0.055, HR = [1.02, 1.09]) and reaching its smallest value for users with more than six years of tenure (0.044, HR = [1.01, 1.08], $p < .01$). This monotonic decline is clearly visible in Figure 3. The waiting period placebo coefficients (*Received Answer* $\times$ *Post-Question*) remain effectively zero across all strata, ruling out pre-trend confounds. These findings support Hypothesis 2: the generalized reciprocity effect, while present throughout the user lifecycle, is substantially stronger among newcomers and attenuates as users gain experience on the platform. This pattern is consistent with our theoretical account that newcomers, who have not yet internalized community-specific norms,

are most susceptible to the moral and emotional mechanisms that drive reciprocity, while socialized users are increasingly governed by community-specific incentive structures.

5.5 Response Time Moderation (H3)

We hypothesized (H3) that faster responses would strengthen the reciprocity effect, based on the prediction that temporal immediacy intensifies gratitude and reinforces perceptions of community generosity. Table 8 presents the interaction between the treatment effect and log response time, estimated separately by tenure bucket.

Table 8: Response Time Moderation of the Reciprocity Effect

	≤ 1 Week	1 Week - 1 Month	1 - 6 Months	6 - 12 Months	1 - 3 Years
<i>Base Treatment Effect</i>					
Received Answer × Post-Answer Received	0.0864*** (0.0083)	0.0968*** (0.0133)	0.0987*** (0.0082)	0.0913*** (0.0094)	0.0713*** (0.0094)
<i>Response Time Interaction</i>					
Treatment × log(Response Time)	0.0114*** (0.0034)	-0.0101* (0.0048)	-0.0135*** (0.0029)	-0.0088** (0.0033)	0.0001 (0.0033)
<i>Waiting Period × Response Time</i>					
Post-Question × log(Response Time)	-0.0153***	-0.0181***	-0.0173***	-0.0101**	0.0001
Intervals	4,870,545	1,839,458	4,811,021	3,572,446	7,821,458
Events	74,469	63,928	188,435	153,918	315,435

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

Our results do not support Hypothesis 3 in its original form and instead reveal a more nuanced, heterogeneous pattern across the user experience spectrum.

Newcomers: Faster Answers Slightly Strengthen Reciprocity. For users with less than one week of tenure, the speed interaction coefficient is positive and marginally

statistically significant ($\gamma = 0.013$, $SE = 0.005$, $p < .05$). This indicates that, among newcomers, faster answers are associated with a *weaker* reciprocity effect—the opposite of what our directional hypothesis predicted. However, the magnitude of the moderation is small relative to the substantial base effect observed for this group, and we interpret this as an essentially null moderation: fast and slow answers produce roughly equivalent boosts in helping behavior among newcomers.

Early Experienced Users: Faster Answers Strengthen Reciprocity. Among users with one week to one month of tenure ($\gamma = -0.010$, $SE = 0.005$, $p < .05$) and one to six months of tenure ($\gamma = -0.010$, $SE = 0.004$, $p < .05$), the speed interaction is negative, indicating that faster answers are associated with a stronger reciprocity effect—consistent with our original prediction. The effect for users with six to twelve months of tenure is not statistically significant ($\gamma = -0.004$, $SE = 0.004$, $p > .10$).

Established Users: Slower Answers Increase Reciprocity. For users with greater platform experience, the speed interaction coefficient turns positive and statistically significant. The effect is marginally significant for users in the one-to-three-year range ($\gamma = 0.007$, $SE = 0.004$, $p < .10$) and grows substantially for the most experienced users: three to six years ($\gamma = 0.026$, $SE = 0.005$, $p < .001$) and more than six years ($\gamma = 0.024$, $SE = 0.005$, $p < .001$). This indicates that, contrary to our prediction, longer response times are associated with a *stronger* reciprocity effect among the most experienced users.

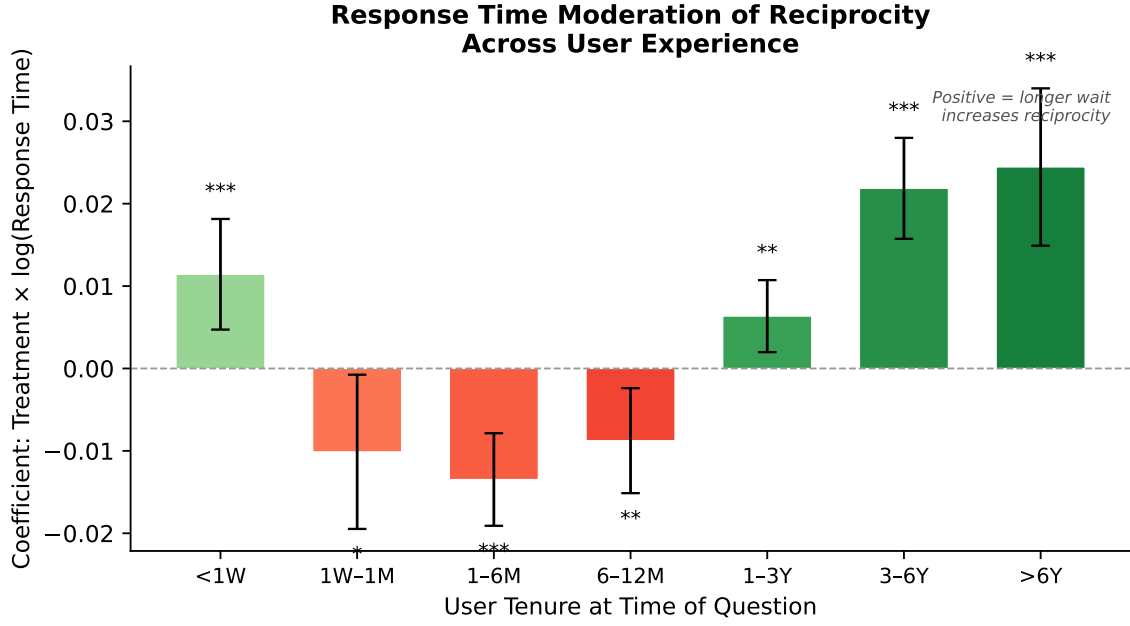


Figure 4: Response Time Moderation Across User Experience. Bars show the interaction coefficient between the treatment effect and log response time, from separate Cox models by tenure bucket. Positive values indicate that longer response times are associated with a stronger reciprocity effect. Error bars indicate 95% confidence intervals.

A potential concern with the response time moderation is that longer response times may proxy for more complex or difficult questions, and complexity rather than delay could drive the differential effect. Our diff-in-diff design mitigates this concern: because we control for the interaction between post-question and response time, we absorb any correlation between the baseline propensity to help of a user and the response time to their questions.

5.6 The Session-Exit Mechanism

We interpret the counterintuitive response time finding through what we term the *session-exit mechanism*. The key insight is not that delayed answers bring disengaged users back to the platform, but rather that fast answers allow users to resolve their immediate need and *leave*. When an experienced user receives a quick answer, their session goal is satisfied; they exit the platform before encountering other questions they might answer. When no answer arrives promptly, the user

remains on the platform while they wait, browsing the site, encountering unanswered questions, and helping others during that extended session. The elevated helping we observe among experienced users after long response times is thus a structural consequence of *sustained presence* during the waiting period, not of a motivational boost triggered by the eventual answer.

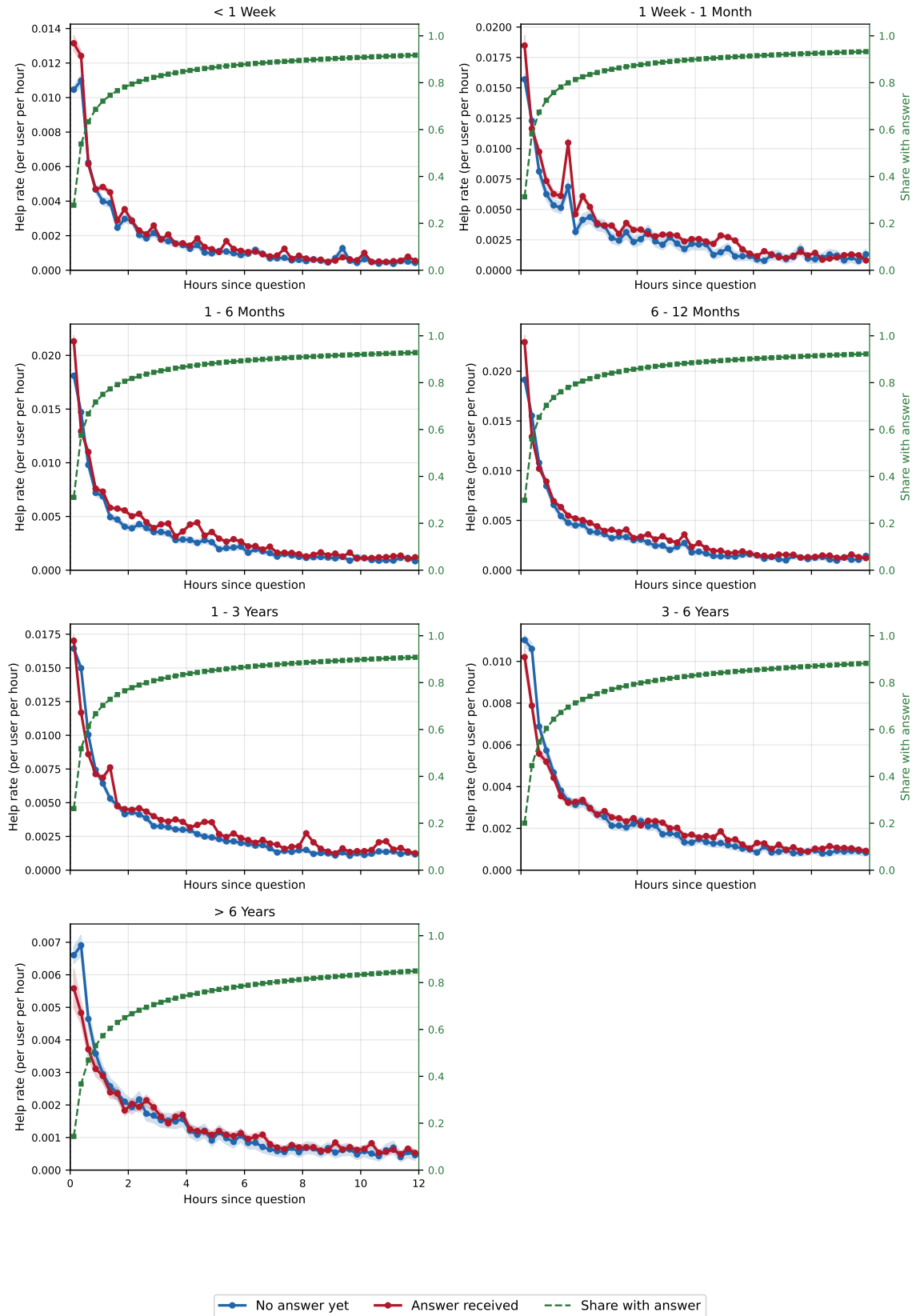


Figure 5: Help rate during the post-question window, separately for users who have not yet received an answer (“No answer yet,” blue) and users who have already received one (“Answer received,” red), by tenure bucket. The green dashed line shows the cumulative share of users who have received an answer by each hour. For experienced users (all panels except < 1 Week), the help rate of users still waiting (blue) consistently exceeds the help rate of users who have already received an answer (red) up to approximately 1–2 hours post-question. This pattern indicates that receiving an answer is associated with platform *exit* rather than continued engagement: users who get a quick answer resolve their need and leave, while users still waiting remain on the platform and are more likely to help others.

Figure 5 provides direct descriptive evidence for this mechanism. For every experienced-user tenure bucket, the help rate of users who have not yet received an answer (blue line) is higher than the help rate of users who have already received one (red line) throughout most of the post-question window—with the two lines crossing only at around one to two hours. This reversal of the expected pattern is incompatible with the emotional immediacy account, which would predict that users who receive an answer quickly should show the highest subsequent helping. Instead, it suggests that receiving an answer is associated with platform *exit*: users whose immediate need is resolved leave the platform, while users still waiting are more likely to remain present and encounter opportunities to help.

The session-exit mechanism predicts that longer response times should be associated with more helping simply because longer waits keep users on the platform longer. This is precisely what the response time interaction captures in the Cox models: for experienced users, each standard deviation increase in log response time is associated with a larger treatment effect on helping hazard. This is not because slow answers motivate stronger reciprocity emotionally, but because slow answers preserve the conditions—continued platform presence—under which helping can occur.

This interpretation is further supported by several patterns in the Cox model estimates. First, the base treatment effect in the speed-interaction model (Model B) is revealing. Because all covariates are mean-centered before fitting, the base coefficient for *isTreatedActive* captures the treatment effect evaluated at the sample mean of the (standardized) response-time interaction term—that is, at the average response time in the data. For several experienced-user buckets, this mean-centered base effect is small or near zero (e.g., $b = -0.006$, $p < .05$ for 1–3 years), indicating that at the average response time, experienced users show little net reciprocity boost. The effect emerges almost entirely from the positive response time interaction, consistent with the session-exit account. Second, the waiting period $\times$ response time interaction is positive and significant across all buckets ($b = 0.015$

to 0.061, all $p < .001$), indicating that longer response times are associated with greater helping activity *during* the waiting period as well—precisely the pattern expected if users are helping while they wait, rather than after they receive a notification. We note that while the session-exit mechanism is our primary interpretation, complementary psychological processes—such as delayed reflection, or the satisfaction of eventually receiving a thorough answer—may also contribute and cannot be definitively ruled out from the present data alone.

5.7 Robustness Checks

We conducted several analyses to assess the robustness of our main findings.

Waiting Period Diagnostics. As reported above, the waiting period coefficient (*treatedPostQuestion*) is significant for newcomers, indicating that treated users are already more active between asking and receiving an answer. In case the matching was imperfect, which we might not detect based on the observable characteristics and matching diagnostics, our diff-in-diff design still identifies an additional lift at the point of receiving the answer for the treatment group. Users can likely gauge the likelihood of receiving an answer because they can see the views and votes their question accumulates, as well as the comments that make suggestions or ask for clarification. These signs of engagement are more likely for answers that receive an answer and might keep the user on the platform. This is also supported by the fact that we observe this pre-treatment difference across all tenure strata. We therefore interpret this as reflecting active-session dynamics rather than a failure of the parallel trends assumption. The lift at the time of receiving an answer stands nonetheless for newcomers.

6 Discussion

6.1 Summary of Findings

We sought to address the empirical challenges that have limited our understanding of generalized reciprocity on online platforms. By developing a matched difference-in-differences survival analysis applied to over 21 million questions on Stack Overflow, we provide evidence that receiving help causally increases a user’s propensity to help others—confirming the basic prediction of generalized reciprocity in a large-scale field setting with substantially stronger identification than prior work. We show that this effect is concentrated among newcomers and declines sharply with platform experience, supporting the view that reciprocity functions primarily as a recruitment mechanism rather than a driver of sustained contribution. Our analysis of response time yields a counterintuitive finding: rather than amplifying reciprocity through emotional immediacy, longer response times are associated with stronger effects among experienced users, a pattern we attribute primarily to a session-exit mechanism whereby fast answers resolve the user’s immediate need and prompt platform exit before they encounter opportunities to help.

6.2 Contributions to the Measurement of Reciprocity

Our primary contribution is methodological. We demonstrate that prior estimates of generalized reciprocity in online communities were likely upward-biased due to the confounding of reciprocity with baseline user activity. Our matched survival analysis addresses this problem through three design elements: propensity score matching across users on tenure, activity, and question characteristics ensures comparability between observations that do and do not receive answers; the difference-in-differences structure—comparing within-observation changes before and after the answer, across treatment groups—isolates the causal effect of receiving help from stable individual differences; and the Cox hazard framework accommodates

the continuous-time nature of helping behavior and the time-varying treatment.

We note that our design does not employ user fixed effects, which might appear to be a natural alternative for controlling unobserved heterogeneity. Because our matching is across different users (not within-user repeated observations), and because the Cox model already exploits within-observation temporal variation through its time-varying covariate structure, user fixed effects are neither appropriate nor necessary in this framework. The propensity score matching absorbs between-user differences in observable characteristics, while the difference-in-differences structure within each matched pair addresses unobserved stable differences. Prior studies that observed positive correlations between help received and help given without such controls likely captured variation in user type rather than the causal effect of reciprocity (??). Our approach can serve as a template for future studies seeking to measure reciprocity from observational platform data.

6.3 Reciprocity as a Recruitment Mechanism

Our finding that reciprocity is strongest among newcomers has important theoretical implications. It suggests that the role of generalized reciprocity in online communities is not to sustain cooperation broadly but to solve a more specific problem: recruiting new contributors. First contributions are the most difficult for platforms to encourage (??), and our results show that reciprocity provides a powerful initial push. The moral obligation to reciprocate operates precisely when community-specific incentives are not yet effective—before newcomers have learned the norms or developed a sense of group identity.

This perspective helps reconcile the mixed findings in the prior literature. Studies that focus on aggregate contributions may underestimate the importance of reciprocity because it drives a relatively small share of total helping activity. But if reciprocity is concentrated among newcomers, its contribution to the community's long-term health may be disproportionate to its aggregate share: each newcomer

recruited through reciprocity may go on to become a sustained contributor motivated by other mechanisms.

6.4 The Limits of Emotional Immediacy in the Field

Perhaps our most provocative finding concerns response time. Laboratory research on gratitude and reciprocity emphasizes the role of emotional immediacy: the closer in time a benefit is received to the act of helping, the stronger the emotional response and the more likely a reciprocal action (???). Our results challenge the transportability of this prediction to field settings.

For newcomers, response time is irrelevant—the reciprocity effect is equally strong whether the answer arrives in minutes or hours. This null result is itself informative. It suggests that among newcomers, the moral obligation to reciprocate is sufficiently robust that its activation does not depend on the intensity of the emotional trigger. The logic of universalization—“I should help because I was helped”—operates as a binary norm that is either activated or not, rather than a continuous function of emotional intensity.

For experienced users, the positive relationship between response time and the reciprocity effect contradicts the emotional immediacy account entirely. As described in our results, we propose that this primarily reflects the *session-exit mechanism*: experienced users whose need is resolved quickly by a fast answer exit the platform before encountering other questions they could answer. Users who wait longer remain on the platform during the waiting period and are more likely to help others while there. The elevated helping associated with longer response times is thus largely a structural consequence of *sustained presence* during the waiting session, rather than a motivational boost triggered by the answer itself. Direct descriptive evidence for this mechanism is visible in Figure 5: for experienced users across all tenure levels, the help rate of users still waiting for an answer consistently exceeds the help rate of users who have already received one, up to

approximately one to two hours post-question—the opposite of what emotional immediacy would predict. We acknowledge that we cannot definitively rule out complementary psychological processes, such as delayed reflection or the satisfaction of receiving a thorough answer, contributing alongside the session-exit channel.

This finding highlights a broader lesson for translating laboratory theories to field settings. In controlled experiments, participants are present throughout the interaction; there is no platform to leave, no session that ends when a need is met. In field settings, the behavioral dynamics of platform engagement—visit initiation, session duration, exit timing—interact with psychological mechanisms in ways that controlled experiments cannot capture. The session-exit mechanism is not an anomaly but a predictable consequence of how users behave on episodic, task-driven platforms: they arrive with a goal, and when the goal is met quickly, they leave. Future theorizing about reciprocity in online communities should incorporate these structural features of platform engagement alongside the emotional and moral mechanisms that have been the traditional focus.

6.5 Practical Implications

Our results offer guidance for platform designers, though we note that the landscape of knowledge-sharing platforms is evolving rapidly and that the specific dynamics we document on Stack Overflow may manifest differently on other platforms.

First, because reciprocity is most powerful for newcomers, platforms should prioritize ensuring that newcomers' questions receive helpful answers. Questions from new users could be given visibility boosts in recommendation algorithms, and experienced users could be nudged to answer questions posed by newcomers.

Second, our response time findings suggest that for newcomers, the *fact* of receiving an answer matters more than how quickly it arrives. Platforms need not

sacrifice answer quality for speed when serving newcomers; what matters is that they receive help at all.

Third, the session-exit mechanism suggests a subtle design tension: fast answers, while clearly valuable for question-askers, may reduce helping by experienced users by ending their platform session prematurely. Platforms could partially offset this effect by surfacing unanswered questions to experienced users at the moment they receive their answer—capitalizing on the brief window before they exit to create an opportunity for helping. This would not delay answers (which is undesirable) but would extend the session slightly beyond the need-resolution moment.

Fourth, a growing challenge for knowledge-sharing platforms is the emergence of generative AI models that provide direct assistance to users. If users increasingly receive help from AI rather than from community members, the human-to-human interactions that trigger reciprocity may diminish. Since our findings indicate that reciprocity is crucial for recruiting newcomers, this shift may pose particular challenges for sustaining the pipeline of new contributors. However, the core mechanism we identify—that receiving help motivates paying it forward—may generalize to other settings where human-to-human interaction remains central, such as mentorship platforms, internal knowledge bases, and community forums in domains where AI assistance is less developed. Platforms may need to develop strategies to preserve or redirect the reciprocal impulse in an AI-mediated environment, though the effectiveness of such strategies remains an open empirical question.

6.6 Limitations and Future Research

Our study has several limitations that suggest directions for future research.

First, we adopted an outcome-based approach, measuring reciprocity through observed helping behavior. While this provides high external validity, it limits

our ability to identify the precise psychological mechanisms at work. We cannot directly observe whether users help out of gratitude, moral obligation, or because remaining on the platform while waiting for an answer put them in the right place and time to encounter other questions. Future research combining behavioral data with in-the-moment experience sampling could help disentangle these mechanisms.

Second, our identification strategy relies on the assumption that, conditional on the matching covariates and the difference-in-differences structure, receiving an answer is as good as being randomly assigned. The propensity score matching—including the tag-level answer rate, which addresses question difficulty—substantially improves comparability, and all post-matching standardized mean differences fall below 0.03. However, unobserved time-varying factors—such as changes in a user’s workload, interest, or the availability of alternative help sources—could in principle bias our estimates.

Third, while we match on the tag-level answer rate to mitigate the concern that response time proxies for question complexity, within-tag variation in difficulty is not fully captured by this aggregate measure. Future work could exploit more granular measures of question complexity—such as text-based difficulty scores or the number of prior unanswered questions in the same topic—to further strengthen the identification of the response time effect.

Fourth, our response time findings are consistent with the session-exit mechanism, but we measure this indirectly through the moderating effect of response time on helping behavior, and through the descriptive pattern in Figure 5. Future research with access to fine-grained session data—precise login and logout timestamps, page-view sequences, or inactivity-based session boundaries—could directly test whether fast answers shorten sessions and whether the additional helping associated with longer waits occurs within continuous sessions or across separate visits.

Fifth, we focus on Stack Overflow, a large, open, and relatively anonymous

platform whose activity has declined in recent years. The core question-and-answer dynamics we study are common to many knowledge-sharing platforms, but the specific magnitudes and patterns may differ in smaller, more intimate communities, in intra-organizational settings where users know each other personally, or on platforms with different incentive structures. Future studies should assess the generalizability of our findings across diverse platform types.

Finally, while we demonstrate that reciprocity declines with experience, we do not fully characterize what replaces it. Our theoretical framework suggests that community-specific norms and status motives take over, but a complete account of the motivational trajectory would require integrating our reciprocity analysis with measures of these alternative mechanisms.

7 Conclusion

Generalized reciprocity occupies a central place in theories of online cooperation, yet measuring it credibly in field settings has proven elusive. We developed a matched difference-in-differences survival analysis that addresses the endogeneity, selection, and temporal limitations that have plagued prior work and applied it to over 21 million questions on Stack Overflow. Our findings confirm that generalized reciprocity operates in this setting but reveal that its role is more circumscribed than previously assumed: it is primarily a mechanism for recruiting new contributors, whose susceptibility to the moral and emotional triggers of reciprocity has not yet been displaced by community-specific norms. Our analysis of response time reveals a further nuance. Contrary to laboratory-based predictions, the speed of help does not amplify reciprocity for newcomers and, among experienced users, longer response times actually produce stronger effects—a pattern we attribute primarily to a session-exit mechanism: fast answers resolve experienced users’ immediate need and prompt them to leave the platform, while slower answers keep users present longer, creating the occasions for helping that produce

the apparent reciprocity boost. These findings underscore the importance of developing measurement strategies that match the complexity of the field settings in which reciprocity operates, and they highlight the need for theories that integrate psychological mechanisms with the structural dynamics of platform engagement.

A Matching Diagnostics

Figure 6 shows that there is sufficient common support between the treatment and the control group. The propensity score distribution between the original treated and control groups is similar. The distribution of the matched control is nearly identical to the treated group.



Figure 6: Relationship between user experience and strength of status pursuit

Figure 7 displays the standardized mean differences (SMD) between the matching variables before and after matching. While there are noticeable differences before the matching, all SMDs are very close to zero after the matching.



Figure 7: Visualization of balance in the unmatched and matched dataset

B Help Rate Over the Observation Window by Tenure

C Analysis of Life-Time Contributions

What remains unclear is whether users who become contributors through generalized reciprocity continue to contribute consistently and become valuable commu-

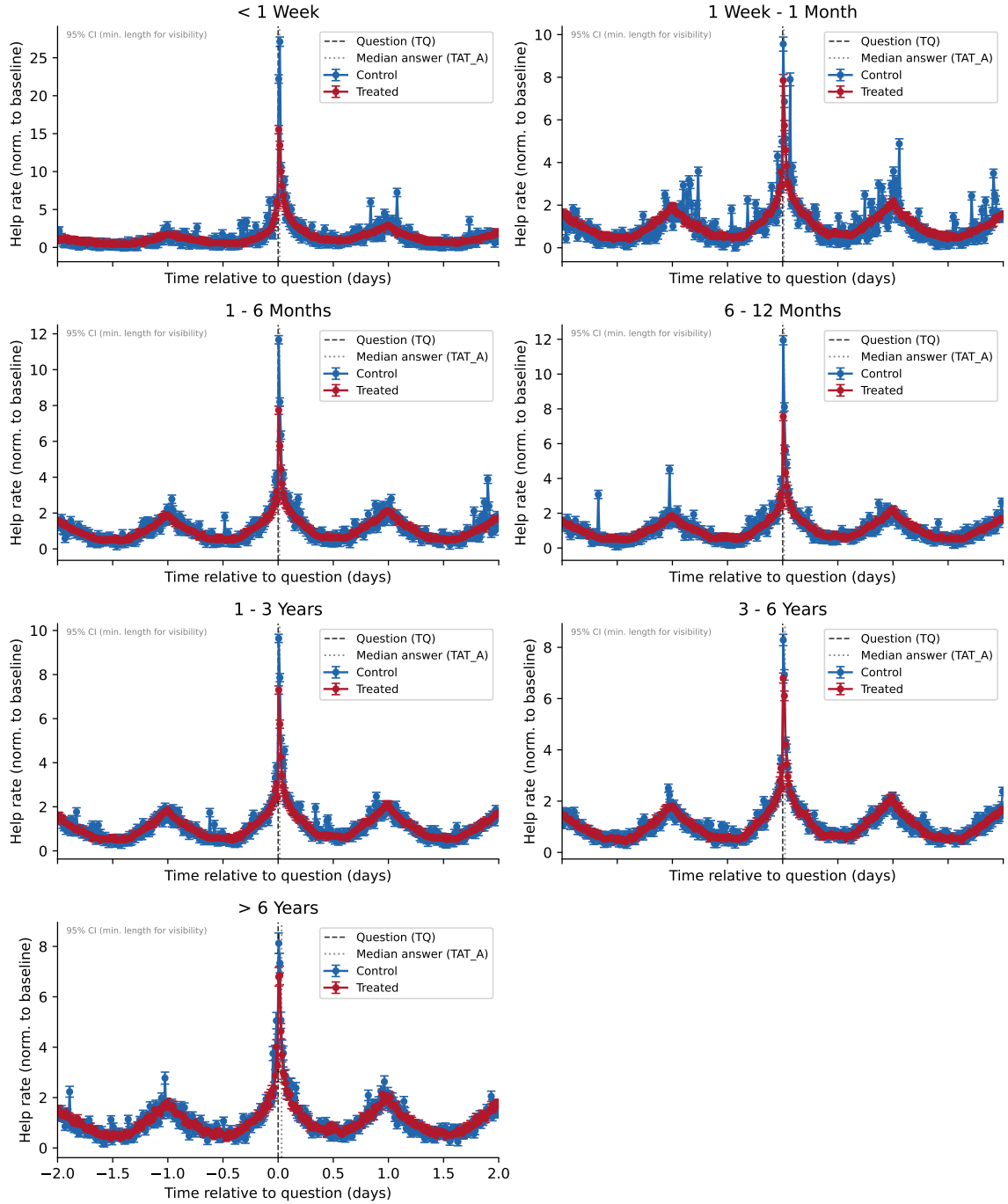


Figure 8: Help Rate Over the Observation Window, by Tenure Bucket. Each panel plots the normalized help rate (relative to each group’s pre-question baseline) for treated users (red) and matched control users (blue) over the ± 2 -day window around the question-posting time (T_Q , dashed vertical line). The dotted vertical line marks the median answer arrival time (T_A). Error bars indicate 95% confidence intervals. The post-question lift in treated users’ help rate is visible across all tenure buckets and is most pronounced among newcomers (tenure < 1 week), consistent with the pattern reported in Table 7 and Figure 3.

nity members. To explore this, I compare the contributions of reciprocity-activated users with those of all contributors selected for Study 1. Specifically, I examine the number of times users from each group provided and received help up to July 14, 2024. The results are presented in Table 9. Across all quartiles and at the median, reciprocity-activated users ask and provide more help than their counterparts. For example, at the median, reciprocity-activated users provide eight answers and ask 13 questions, compared to six answers and 11 questions among all contributors. Similarly, the 75th percentile of reciprocity-activated users provides 23 answers and asks 29 questions, compared to 19 answers and 22 questions for all contributors. These findings suggest that users who begin contributing after receiving help become equally, if not more, engaged in the community’s helping dynamics compared to all contributors. Furthermore, this analysis demonstrates that once users start contributing, the act of giving and receiving help becomes (at least temporarily) decoupled.

Table 9: Statistics on Contributions by User Sample

	Answers Provided			Questions Asked		
	Q_1	Med	Q_3	Q_1	Med	Q_3
Reciprocity-activated ($N = 30,160$)	3.0	8.0	23.0	7.0	13.0	29.0
All Contributors ($N = 287,959$)	2.0	6.0	19.0	6.0	11.0	22.0